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# MATHEMATICS

# MA01

(9660/MA01) Unit P1 Pure Mathematics

Mark scheme

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*20Ixma01/MS*

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**Key to mark scheme abbreviations**

| | |
|---|---|
| **M** | Mark is for method |
| **m** | Mark is dependent on one or more M marks and is for method |
| **A** | Mark is dependent on M or m marks and is for accuracy |
| **B** | Mark is independent of M or m marks and is for method and accuracy |
| **E** | Mark is for explanation |
| **√or ft** | Follow through from previous incorrect result |
| **CAO** | Correct answer only |
| **CSO** | Correct solution only |
| **AWFW** | Anything which falls within |
| **AWRT** | Anything which rounds to |
| **ACF** | Any correct form |
| **AG** | Answer given |
| **SC** | Special case |
| **oe** | Or equivalent |
| **A2, 1** | 2 or 1 (or 0) accuracy marks |
| **–*x* EE** | Deduct *x* marks for each error |
| **NMS** | No method shown |
| **PI** | Possibly implied |
| **SCA** | Substantially correct approach |
| **sf** | Significant figure(s) |
| **dp** | Decimal place(s) |

| Q | Answer | Marks | Comments |
|---|---|---|---|
| 1(a)(i) | $\frac{5}{2}$ | B1 | |
| 1(a)(ii) | 3 | B1 | |
| 1(b) | $\frac{dy}{dx} = \frac{5}{2} \times 8x^{\frac{3}{2}} - 3 \times 2x^2$ | M1 | oe<br>ft $p$ and $q$ from part **(a).**<br>Must use coefficients 8 and 2 at this stage.<br>One correct term, simplified or unsimplified. |
| | $\frac{dy}{dx} = 20x^{\frac{3}{2}} - 6x^2$ | A1 | CAO |
| | **Total** | **4** | |

| Q | Answer | Marks | Comments |
|---|---|---|---|
| 2(a) | $(x-5)^2$<br>or<br>$-\frac{b}{2a} = 5$ or $-\frac{b}{2} = 5$<br>or<br>$x^2 - 10x + 25$<br>or<br>Indicates there is a stationary point at $(5, 0)$ | M1 | |
| | $b = -10 \quad c = 25$ | A1 | CSO. Both values.<br>Correct $b$ and $c$ but NMS scores 2/2. |
| 2(b) | Correctly orientated symmetrical quadratic parabola tangential to $x$-axis or symmetrical about the line $x = 5$ | B1 | |
| | $(0, 25)$ labelled on $y$-axis. | B1ft | ft their $c$<br>Condone label given as the $y$-value only. |
| | Correctly positioned vertex labelled as $(5, 0)$ | B1 | Accept correctly positioned vertex with 5 indicated or implied on the $x$-axis. |
| | **Total** | **5** | |

| Q | Answer | Marks | Comments |
|---|---|---|---|
| 3(a)(i) | $y=\frac{3}{2}x+3$ | M1 | Attempt at $y=mx+c$ or Attempt to find gradient using two correct points on the line. |
| | (Gradient =) $\frac{3}{2}$ | A1 | oe |
| 3(a)(ii) | ( $y$ -coordinate =) 3 | B1 | Condone given as coordinates. |
| 3(b)(i) | $(m=)\ -\frac{2}{3}$ | B1ft | ft their gradient for $L_1$. |
| 3(b)(ii) | $0=-\frac{2}{3}x+\frac{22}{3}$ or $\frac{2}{3}x=\frac{22}{3}$ and $(x=)$ 11 | B1 | AG. Zero substituted for $y$ in the equation of $L_2$ and correct answer. |
| 3(c) | $(D=)\left(\frac{11}{2},\frac{3}{2}\right)$ | B1ft | ft their $y$ -coordinate of $B$. PI in later working. |
| | $y-\frac{3}{2}=\text{their gradient}\times\left(x-\frac{11}{2}\right)$ | M1 | ft their gradient of $L_1$. May see $2y-3x=k$ and substitution of coordinates of $D$ to find $k$ . But must be full method. |
| | $y-\frac{3}{2}=\frac{3}{2}\left(x-\frac{11}{2}\right)$ | A1 | CAO. oe |
| | **Total** | **8** | |

| Q | Answer | Marks | Comments |
|---|---|---|---|
| 4(a) | $(2-x)^4 =$ $\left[(2)^4\right]+\left[4(2)^3(-x)\right]+6(2)^2(-x)^2+4(2)(-x)^3+\left[(-x)^4\right]$ | M1 | For either (1), [4], 6, 4, (1) oe unsimplified. or $\binom{4}{2}(2)^2(-x)^2$ or $\binom{4}{3}(2)(-x)^3$ oe, $x$ not needed, PI. |
| | $a=24$ | A1 | If correct working for $a$ seen condone 24 seen in expansion but not explicitly stated. |
| | $b=8$ | A1 | $a$ not shown but correct $b$ seen but NMS merits M1A0A1. Condone $24x^2$ or $8x^3$. |
| 4(b) | $\left(\left(2-\frac{1}{\sqrt{2}}\right)^4 =\right)$ $16-32\left(\frac{1}{\sqrt{2}}\right)+24\left(\frac{1}{\sqrt{2}}\right)^2-8\left(\frac{1}{\sqrt{2}}\right)^3+\left(\frac{1}{\sqrt{2}}\right)^4$ | M1 | Substitution of $x=\frac{1}{\sqrt{2}}$ into their expansion. |
| | $16-\frac{32}{\sqrt{2}}+12-\frac{4}{\sqrt{2}}+\frac{1}{4}$ | M1 | Powers correctly evaluated in an expression. |
| | $16-16\sqrt{2}+12-2\sqrt{2}+\frac{1}{4}$ | M1 | Denominators rationalised in an unsimplified expression. |
| | $\frac{113-72\sqrt{2}}{4}$ | A1 | Accept other correct forms, e.g. $\frac{226-144\sqrt{2}}{8}$ NMS = 0 |
| | **Total** | **7** | |

| | | | |
|---|---|---|---|
| 4(b) ALT | $\left((2-x)^4 =\right)$ $16-32\left(\frac{\sqrt{2}}{2}\right)+24\left(\frac{\sqrt{2}}{2}\right)^2-8\left(\frac{\sqrt{2}}{2}\right)^3+\left(\frac{\sqrt{2}}{2}\right)^4$ | M2 | Substitution of $x=\frac{\sqrt{2}}{2}$ into their expansion. |
| | $16-16\sqrt{2}+12-2\sqrt{2}+\frac{1}{4}$ | M1 | Powers correctly evaluated in an expression. |
| | $\frac{113-72\sqrt{2}}{4}$ | A1 | Accept other correct forms, e.g. $\frac{226-144\sqrt{2}}{8}$ NMS = 0 |

| Q | Answer | Marks | Comments |
|---|---|---|---|
| 5(a) | $x-15+\frac{14}{x}$ or $x-15+14x^{-1}$ | M1 | Expansion of numerator and division by $x$. Allow error in one term if two other terms correct. |
| | $(y=)\ x-15+14x^{-1}$ | A1 | oe. Correct simplification |
| | $\frac{dy}{dx}=1-14x^{-2}$ | A1ft | oe. ft their rearrangement of $y$ provided their rearrangement contains a term in $x^{-1}$ |
| | $x=2 \Rightarrow \frac{dy}{dx}=1-\frac{14}{4}=-\frac{5}{2}$ | B1ft | ft their $\frac{dy}{dx}$ |
| | $x=2 \Rightarrow y=-6$ | B1 | ft their $-\frac{5}{2}$ provided it came from a correct method. ft their $-6$ |
| | $(y-(-6))=-\frac{5}{2}(x-2)$ | M1ft | May see substitution of $x=2,\ y=-6$ and $m=-\frac{5}{2}$ into $y=mx+c$ but must be a complete method to find $c$. |
| | $(y-(-6))=-\frac{5}{2}(x-2)$ | A1 | ACF. Correct equation simplified or unsimplified. |
| 5(b) | $\frac{dy}{dx}=1-14\times(-4)^{-2}\left[=0.125>0\right]$ | B1ft | ft their $\frac{dy}{dx}$ Substituting $x=-4$ into their $\frac{dy}{dx}$ to show it has a positive value. |
| | Increasing | E1ft | ft. Allow 'decreasing' if their $\frac{dy}{dx}<0$ for their $\frac{dy}{dx}$ Dependent on B1 scored. |
| | **Total** | **9** | |

| Q | Answer | Marks | Comments |
|---|---|---|---|
| 6(a)(i) | Arithmetic (sequence) | E1 | With an attempt at an explanation. |
| | The sequence increases by 22 each time. | E1 | Explanation implying that the difference between each term is constant. |
| 6(a)(ii) | $36+(n-1)d$ or $a+(n-1)22$ | M1 | $a+(n-1)d$ with $a$ or $d$ correct and the other a letter. |
| | $36+(n-1)22$ | A1 | CAO oe |
| 6(b) | $(\text{Total boxes sold}=)$ 7500 | B1 | PI in later working. |
| | $\frac{1}{2}N(2\times 36+(N-1)\times 22)$ | M1 | oe. Expression for total number of boxes sold in $N$ months. Condone other letters used instead of $N$ throughout. |
| | $11N^2+25N-7500=0$ | A1 | oe. Correct quadratic equation in $N$. PI. |
| | $(N=)\ \frac{-25\pm\sqrt{25^2-4\times 11\times(-7500)}}{2\times 11}$ or $(11N+300)(N-25)\ (=0)$ | M1 | PI by 25 and $-27.27...$ oe. Must have both values. Correct use of quadratic formula to solve their quadratic equation, condoning one error in signs or Correct factorisation ignoring signs. |
| | 25 | A1 | Both values given as final answer scores A0 |
| | **Total** | **9** | |

| Q | Answer | Marks | Comments |
|---|---|---|---|
| 7(a) | $\int\left(3x^2+ax-36\right)\mathrm{d}x$ | | |
| | $=x^3+\frac{a}{2}x^2-36x\ \ (+c)$ | M1 | Attempt to integrate $\frac{\mathrm{d}y}{\mathrm{d}x}$ with at least two terms correct. |
| | $\left(y=\right)x^3+\frac{a}{2}x^2-36x\left(+c\right)$ | A1 | oe. |
| | $1+\frac{a}{2}-36+c=-7$ or $27+\frac{9a}{2}-108+c=-5$ | m1 | oe. Substituting either $x=1$ and $y=-7$ or $x=3$ and $y=-5$ into their equation for $y$ in an attempt to find an equation in $a$ and $c$. |
| | $\frac{a}{2}+c=28$ | A1 | oe. For each correct equation simplified or unsimplified. PI. |
| | $\frac{9a}{2}+c=76$ | A1 | |
| | $a=12$ and $c=22$ | B1 | Both correct. PI |
| | $\left(y=\right)\ x^3+6x^2-36x+22$ | A1ft | ft their $a$ and $c$ provided both method marks scored. |
| 7(b)(i) | $\left(\frac{\mathrm{d}^2y}{\mathrm{d}x^2}=\right)\ 6x+12$ | B1ft | ft their $\frac{\mathrm{d}y}{\mathrm{d}x}$ with their $a$ substituted. |
| 7(b)(ii) | $\left(\frac{\mathrm{d}y}{\mathrm{d}x}=\right)\ 3x^2+12x-36=0$ and $\left(x+6\right)\left(x-2\right)=0$ | M1 | oe. Sets their $\frac{\mathrm{d}y}{\mathrm{d}x}$ equal to zero with their $a$ and attempts to solve for $x$. PI by $x=2$ |
| | $x=2\Rightarrow\frac{\mathrm{d}^2y}{\mathrm{d}x^2}=6\times2+12\left[=24>0\right]$ | M1 | Establishes that $\frac{\mathrm{d}^2y}{\mathrm{d}x^2}$ is positive at $x=2$<br>Must be working with correct $\frac{\mathrm{d}^2y}{\mathrm{d}x^2}$ and $x=2$ at this stage. |
| | $x=2$ | A1 | CSO.<br>Failure to show that $\frac{\mathrm{d}^2y}{\mathrm{d}x^2}$ is positive at $x=2$ scores A0. |
| | **Total** | **11** | |

| Q | Answer | Marks | Comments |
|---|---|---|---|
| 8(a)(i) | $\frac{a}{1-r}=425$<br>$a+ar=408$<br>$425(1-r)+425(1-r)r=408$<br>or<br>$\frac{408}{1-r^2}=425$<br>$r=\frac{1}{5}$ | M1<br>A1<br>M1<br>A1 | oe. One correct equation.<br>oe. Both correct equations.<br>Attempts to solve simultaneously by elimination and forms equation in $r$. Simplified or unsimplified.<br>AG. From correct working. |
| 8(a)(ii) | $\frac{a}{1-\frac{1}{5}}=425$<br>or<br>$a+\frac{1}{5}a=408$<br>$(a=)\ 340$ | M1<br>A1 | oe. Substitutes $r=\frac{1}{5}$ into either of their equations in part (a). |
| 8(b) | $\left(\sum_{n=k}^{k+1}u_n=\right)\sum_{n=1}^{k+1}u_n-\sum_{n=1}^{k-1}u_n$<br>or<br>$ar^{k-1}+ar^k$<br>$\frac{340\left(1-\left(\frac{1}{5}\right)^{k+1}\right)}{1-\frac{1}{5}}-\frac{340\left(1-\left(\frac{1}{5}\right)^{k-1}\right)}{1-\frac{1}{5}}$<br>or<br>$340\left(\frac{1}{5}\right)^{k-1}+340\left(\frac{1}{5}\right)^{k}$<br>$425\left(\frac{1}{5}\right)^{k-1}\left(1-\left(\frac{1}{5}\right)^2\right)$<br>or<br>$340\left(\frac{1}{5}\right)^{k-1}\left(1+\frac{1}{5}\right)$<br>$\left(\sum_{n=k}^{k+1}u_n=\right)\ 408\left(\frac{1}{5}\right)^{k-1}$ | B1<br>M1<br>M1<br>A1 | PI in later working.<br>oe. Correct substitution of $r$ and $a$. Simplified or unsimplified. ft their $a$.<br>Simplification and factorisation, unsimplified.<br>CSO |
| | **Total** | **10** | |

| Q | Answer | Marks | Comments |
|---|---|---|---|
| 9(a) | $(y=)\ x^{\frac{2}{3}}-8x^{\frac{1}{3}}+5$ | B1 | Correct expansion simplified or unsimplified. |
| | $\int\left(\left(x^{\frac{1}{3}}-4\right)^2-11\right)\mathrm{d}x=$ <br> $\frac{3}{5}x^{\frac{5}{3}}-6x^{\frac{4}{3}}+5x(+c)$ | B2ft | Simplified or unsimplified. <br> For B2 condone omission of $+c$ <br> Two correct terms only scores B1B0 <br> ft their expansion. |
| 9(b) | $\left(\frac{3}{5}(8)^{\frac{5}{3}}-6(8)^{\frac{4}{3}}+5(8)\right)-\left(\frac{3}{5}(1)^{\frac{5}{3}}-6(1)^{\frac{4}{3}}+5(1)\right)$ | M1ft | Identifies correct definite integral and attempts $\mathrm{F}(8)-\mathrm{F}(1)$. <br> ft their indefinite integral from part **(a)**. |
| | $\frac{-182}{5}$ or $-36.4$ | A1 | oe. CSO <br> PI by correct final answer. |
| | $\frac{182}{5}$ or $36.4$ | A1ft | oe. Correct area <br> ft the value of their definite integral provided it was negative. |
| 9(c)(i) | $\left(\frac{182}{5}\right)+14$ | M1 | oe. Recognises that they need to increase the area in part **(b)** by 14. |
| | $\frac{252}{5}$ or $50.4$ | A1ft | oe. ft their area from part **(b)** provided it was positive. <br> Re-integrating the equation of $C_2$ scores M0A0. |
| 9(c)(ii) | $(y=)\left(x^{\frac{1}{3}}-4\right)^2-13$ | B1 | ACF |
| | **Total** | **9** | |

| Q | Answer | Marks | Comments |
|---|---|---|---|
| **10(a)** | $16(p+3)^2-4(2)(12p+q+12)\ [>0]$ | **M1** | Use of $b^2-4ac$ with $a$, $b$ and $c$ substituted. |
| | $16p^2+96p+144-96p-8q-96(>0)$ | **A1** | oe. Correct expansion. |
| | $16p^2-8q+48>0$<br>$2p^2-q+6>0$ | **A1** | AG. From correct working.<br>Brackets must be seen to be expanded before final answer given.<br>$>0$ must appear before final line. |
| **10(b)** | $2(0)^2+4(p+3)(0)+12p+q+12=32$ | **M1** | Attempts to use $(0,32)$ to find an equation in $p$ and $q$. |
| | $q=20-12p$ | **A1** | oe. |
| | $2p^2-(20-12p)+6\,[>0]$ | **m1** | oe.<br>Uses their equation in $p$ and $q$ to eliminate $q$ in the inequality. |
| | $p=-7$ and $p=1$ | **A1** | Correct critical values. |
| | $p<-7$ or $p>1$ | **A1** | Condone 'or' omitted but do not condone inclusion of 'and'. |
| | **Total** | **8** | |